

Hook Paper — Opening Spec (for Lyra)

Keeper, 2026-07-26. The opening of the Tegmark hook package's small paper. Casey's structure ("write it like a movie — short action scene, tagline, the α hook"), with the tier-honesty guardrails and the corrected Wyler history. This is the OPENING (~the first page); Lyra develops the rest.

The tagline (the literal opening line)

"What is the smallest object that can do physics?" It is a *question*, not a claim — minimal, AC(0), and exactly what a Tegmark-class reader would ask. Everything opens from it.

The beat structure (movie open — fast, each beat lands)

1. The question (the tagline) — if the universe is a mathematical structure, it should be the *smallest* one that carries the job.
2. The assertion / action scene (stated as the guess it was, "the guess is the story"): QM is 2D, GR is 3D; smallest closed 2D = a circle; circles tile a sphere = smallest closed 3D with a volume; a 1D channel carries the information, capacity 137. (*Plant, don't explain: the original picture was timeless — time entered later as the commitment rate. One clause, a forward hook, not a digression.*)
3. The Wyler payoff — $137 = 1/\alpha$; the number Wyler derived from a bounded symmetric domain (1969), and the guess reproduces it and says *why* ($137 =$ the channel's capacity, read off the geometry).
4. The honest-surprise beat — computed m_p/m_e expecting the inconsistency that ends most such stories; got $6\pi^5 = 1836.12$ vs 1836.15 measured; never found it; six hundred numbers later still haven't. (*The EXPECTATION of failure is the credibility — keep it.*)
5. "Run it yourself" — `python3 verify_bst.py`, 50 predictions in 3 seconds. "Physics fits on GitHub."
6. The ironic MUH reveal (the closing turn) — only afterward did I recognize the object behind the guess is a single *forced, unique* bounded symmetric domain, D_{IV}^5 — not one structure among Tegmark's many, but the one the constraints select. I had not set out to build a foundation for the MUH; the smallest-object question walked me into it.

The rendered draft (shoot at it)

What is the smallest object that can do physics?

Not a theory of everything — a question. If the universe is at bottom a mathematical structure, it ought to be the *smallest* one that can carry the job. So: what is the least thing that can hold quantum mechanics, general relativity, and the exchange of information between them?

Here is the guess this paper started from, stated the way it was first said, because the guess is the story. Quantum mechanics is two-dimensional; general relativity is three. The smallest closed two-dimensional object is a circle. Circles tile a sphere — the smallest closed three-dimensional object that bounds a volume. Between them runs a one-dimensional channel carrying the information, with a finite capacity. Work out that capacity and it is 137. (*The original picture was timeless; where time enters — as the rate at which the substrate commits — came later, and is Section [N].*)

That number is not chosen to impress. $137 = 1/\alpha$, the fine-structure constant — the number Armand Wyler extracted from the volume of a bounded symmetric domain in 1969. The guess reproduces Wyler's result and supplies the thing his critics said it lacked: a reason. 137 is the channel's capacity, read off the geometry, not a radius set to one by hand.

So we computed the next thing — the proton-to-electron mass ratio — expecting the inconsistency that ends most such stories. It came out $6\pi^5 = 1836.12$. Measured: 1836.15. We never found the inconsistency. Six hundred numbers later, we still haven't.

You don't have to take that on faith, and that is the point. The whole thing is a public repository you can run: `python3 verify_bst.py` checks fifty predictions against measurement in three seconds. Physics, it turns out, fits on GitHub.

And only afterward did I recognize what the object behind the guess actually is. The configuration space those choices force is a single, unique bounded symmetric domain — D_{IV}^5 — not one mathematical structure among many, but the one the constraints *select*. I had not set out to build a foundation for the Mathematical Universe Hypothesis. The smallest-object question walked me into it.

The Wyler lineage — CORRECTED from the archive (use THIS; it's stronger)

The remembered version (“Robertson killed it with a dismissive letter, correct work ignored for status”) is not quite what the record says — and the accurate version is *better for us*, because BST answers the recorded objection. - Wyler (Swiss mathematician), 1969: first exact formula for α from the Euclidean volumes of bounded symmetric domains, tied to the invariance group $O(n,2)$ of the wave equation. Agreed with experiment to ± 1.5 ppm. - Freeman Dyson invited him to the Institute for Advanced Study (Princeton), 9/1971–6/1972. It was taken seriously by a great physicist. - Gloria Lubkin, “A Mathematician’s Version of the Fine-Structure Constant,” *Physics Today* 24(8), 1971 — the public attention. - The critique — Robertson, *Phys. Rev. Lett.* 27, 1545 (1971) — and this is the key: it was specific and fair, not a status-dismissal. Robertson’s objection was that Wyler’s expression matches experiment only if the radius of the spaces is arbitrarily set to 1, with no physical reason given (the relation to the invariance group is radius-independent). AND Robertson wrote that α “might be derivable theoretically” and that “Wyler’s number appears to have better chances to be derived from a theory than any of the other numbers that also agree with experiment.” The critic thought it was the *most promising* such number. - Why it was dismissed: not because it was shown wrong — because Wyler could not put it in a convincing physical context / justify the normalization. The community then filed it under numerology. - ★ THE FRAME FOR US: BST supplies exactly the missing piece Robertson named. The domain is not chosen — it is forced (the unique D_{IV}^5), and the normalization is its intrinsic Bergman / Faraut–Korányi measure, not a radius set to one by hand. Robertson’s specific, recorded objection — *why radius = 1?* — is answered by *because the geometry is forced and carries its own measure*. We are not reviving a dismissed curiosity; we are closing the exact gap the referee flagged, fifty-plus years on. That is a cleaner and more honest hook than “ignored for sociology.” - (*Primary-source check before publishing: confirm Robertson’s initials/identity and quote from PRL 27, 1545 directly; the critique-content above is from the secondary record and is consistent, but quote the primary.*)

Tier-honesty guardrails (with this reader, honesty IS the credibility)

1. Keep the intuition and the rigor on different lines. The circle-tiles-a-sphere story is the *journey* (label it a guess); “ D_{IV}^5 is forced/unique” is the *claim* (the uniqueness argument does the work). Draw the seam yourself so the reader doesn’t have to hunt for it.
2. “Reproduces Wyler / supplies the reason” is the honest verb for α — it is an IDENTIFICATION (the integer 137 forced, the exact value matched), not a proof of α . Do not upgrade it.
3. “Six hundred numbers, no inconsistency” is a consistency claim, not “all derived” — it is true and it is a hook; keep it about consistency, not derivation.
4. Plant time, don’t explain it (beat 2’s parenthetical) — the commitment-rate mechanism is a payoff three sections in, not opening material.
5. The scope page (BST_What_We_Claim_And_Do_Not...) is the companion — the opening seduces, the scope page keeps it honest; both ship together.

— Keeper, 2026-07-26. For Lyra (hook-paper lead). Companion: the scope spine (Keeper), the curated toys (Elie), the tier-ledger (Grace). Cal gate before Tegmark; nothing sent without Casey’s direction.